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RESEARCH ARTICLE

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Reducing the incidence of biased algorithmic decisions through feature importance transparency: an empirical study

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ABSTRACT

As firms move towards data-driven decision-making using algorithmic systems, concerns are raised regarding the lack of transparency of these systems which could have ramifications related to users' trust and the potential for provoking discriminatory decisions. Although previous research has developed methods to improve algorithmic transparency, little empirical evidence exists regarding the extent of the effectiveness of these approaches. Drawing upon Rest's theory of ethical decision-making and the literature on algorithmic transparency and bias, we investigate the effectiveness of feature importance (FI), a common transparency-enhancing approach, which illustrates the nature and the weights of the features utilised by an algorithm. Through an online experiment employing a fictitious tool that provided recommendations for selecting employees for a promotion-related training programme, we find that FI is effective when biased recommendations include direct discrimination (i.e. when individuals are treated less favourably on protected grounds such as gender); but is of little assistance when discrimination is indirect (i.e. when a criterion or practice that is apparently neutral, disadvantages a group of individuals who are of a protected class). Additionally, we propose a new transparency approach, using aggregated demographic information, to accompany FI in indirect discrimination circumstances and report the results of testing its effects.

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Algorithmic bias; fairness; ethics; transparency; trust

1. Introduction

With the rapid growth of data analytics, big data technologies, and artificial intelligence, firms are focusing significant investments on algorithmic decision aids, trusting their capabilities with the aim of deriving important insights, improving decision-making, and ultimately gaining a competitive edge (Akter et al., 2021; Davenport et al., 2020; Mikalef et al., 2020). However, despite the increasing adoption of algorithmic decision aids, researchers (e.g., Marabelli & Chan, 2024; Mehrabi et al., 2021; Mikalef et al., 2022; Tsamados et al., 2022) and practitioners (e.g., Blier, 2019; Manyika et al., 2019) have increasingly warned about the detrimental effects of biased algorithmic outputs (Akter et al., 2021; Friedman & Nissenbaum, 2022), which can arise due to technical (e.g., unbalanced data sets) and social (e.g., historical biases against underprivileged communities such as racial minorities) reasons (Dolata et al., 2022). The literature is rife with examples of such biases in the recommendations and decisions made by these systems or by people relying on them in a variety of contexts, such as university admissions (Hao, 2020), visa and immigration decisions (McDonald, 2019), hiring (Dastin, 2022; Weston, 2021), and recidivism prediction (Courtland, 2018; C. Wang et al., 2023).

To raise awareness about the possible dangers of using algorithmic systems as decision aids and to hold firms accountable for their algorithmic-based decisions, various government agencies – such as the U.S. Equal Employment Opportunity Commission and the European Commission – have issued pertinent acts and documents. In addition to improving compliance with anti-discrimination laws, enhancing fairness in relation to algorithmic decisions made in organisations is important in terms of social responsibility and accountability (De-Arteaga et al., 2022). Algorithmic practices that disregard fairness and responsible decision-making cause negative societal consequences and can endanger firms' reputation, performance, competitiveness, and even survival (De-Arteaga et al., 2022; Leicht-Deobald et al., 2019). In fact, the last few years have witnessed several litigation cases in relation to firms' discriminatory decisions and practices due to the use of algorithmic systems (For examples, see Black, 2022; Chadha, 2024; Wiessner, 2024).

Due to the existing limitations of algorithmic systems in terms of abiding by fairness regulations, they are often employed to work alongside human experts in complex decision-making contexts as opposed to acting autonomously (Adomavicius & Yang, 2022; Luong et al., 2021). Approaches to granting human

decision-makers the ultimate decision-making authority aim to help alleviate the adverse impacts of biased algorithmic outcomes (Hamilton & Davison, 2022). To succeed, these approaches require addressing the issue of transparency of algorithms and their outcomes (Ebrahimi & Hassanein, 2021; Köchling & Wehner, 2020; Tambe et al., 2019) as transparency can “help to reveal areas of potential discrimination that may not be apparent otherwise” (Kirkpatrick, 2016, p. 17). Enhancing transparency in the hope of improving the fairness of algorithmic-based decisions is based on the logic that when a human learns about the inner workings of an algorithm, they not only comprehend those inner workings but can and will identify fairness-related issues and take the necessary steps to correct them (Martin, 2019; Vimalkumar et al., 2021).

While algorithmic transparency may help to facilitate fairness perceptions, it may not always be the case, as humans are subject to bounded rationality and cognitive biases. These cognitive limitations and biases of human decision makers may influence appropriate fairness interpretation of algorithmic outcomes (Adomavicius & Yang, 2022), even with algorithmic transparency. Indeed, the extant literature reports mixed findings regarding the influence of transparency on fairness perceptions with some researchers reporting positive relationships (e.g., Angerschmid et al., 2022; Weith & Matt, 2023) and some reporting no impact (e.g., Newman et al., 2020). Interestingly, with few exceptions (e.g., M. K. Lee et al., 2019; R. Wang et al., 2020), the majority of the existing studies on the impact of algorithmic transparency on fairness perceptions are conducted in the context of non-biased algorithmic recommendations. We caution against generalising these algorithmic transparency outcomes to the context of biased algorithmic recommendations. Further, the extant literature has mainly focused on fairness perceptions of algorithmic recommendations. While understanding such fairness perceptions is important, it is the intention to approve algorithmic-based recommendations, which may be biased, that is of practical concern. Thus, while common wisdom suggests that enhancing transparency facilitates fairness of algorithmic-based decisions, we contend that this needs to be validated for biased/discriminatory algorithmic recommendations and extended to behavioural intention to adopt such recommendations. Therefore, this study seeks to empirically investigate the following research question: *How effective is algorithmic transparency in reducing users’ adoption of discriminatory algorithmic recommendations?*

To address our research question, we draw on Rest’s (1986) theory of ethical behaviour¹ and the bodies of literature on algorithmic bias and algorithmic transparency as well as the literature on organisational fairness. Through an online experiment, we

examine the impact of the presence of algorithmic transparency on users’ recognition of the ethical aspects of the situation at hand when dealing with biased algorithmic recommendations. Such recognition, called moral awareness, can reduce users’ behavioural intention to approve biased recommendations. Specifically, we examine the influence of the presence of transparency in the form of a list of important features used by an algorithmic decision aid and their relative contributions (hereafter called feature importance) to the generation of a set of algorithmic recommendations. We contend that feature importance (FI) transparency is particularly effective in direct discrimination situations that disadvantage someone(s) due to their explicit membership in a particular group (e.g., women). However, providing FI transparency may not be as effective when the bias is in the form of indirect discrimination, where a protected group (e.g., women) is disadvantaged despite recommendations being based on seemingly neutral parameters (e.g., due to biases in the data set). For the latter, we propose a new type of transparency that we call aggregated demographic information (ADI). As a companion to FI, ADI depicts (possible) differences among different demographic groups in terms of each of the important features used by an algorithm to generate recommendations. As such ADI enables users to see differences among different demographic groups in terms of one or more features included in an algorithm’s analysis. Such differences between demographic groups can be a source of discrimination in algorithmic recommendations (Kordzadeh & Ghasemaghaei, 2022).

Our findings contribute theoretically to the algorithmic bias literature, which suffers from a dearth of empirical studies that investigate users’ cognitions and behaviours when facing such biases. In addition, this study paves the way for a better understanding of the importance of providing various types of transparency in conditions where the danger of direct and/or indirect discriminatory algorithmic recommendations exists. Furthermore, the results of this study provide important insights for practitioners interested in reducing instances of biased decisions in their organisations. As such, this study takes an important step towards more ethical use of algorithmic decision aids by helping to reduce the incidence of discriminatory algorithm-based decisions in organisations.

2. Theoretical background

2.1. Causes and forms of bias in the outputs of algorithmic systems

Algorithmic bias is a socio-technical issue (Favaretto et al., 2019), which arises due to various reasons including biased, mislabelled, and/or

unbalanced data sets (Akter et al., 2021; Biswas et al., 2020) as well as methodological issues such as overgeneralisation (i.e., providing generic insights as opposed to being specific to situations and contexts), overfitting (i.e., models becoming too specific to the training data and therefore, failing to generalise to novel data) (Akter et al., 2021; Favaretto et al., 2019), and/or the inclusion of protected variables (e.g., sex) or their proxies (Dwork et al., 2012; Pedreschi et al., 2008). Algorithmic models can inherit social biases during their development, training, and/or deployment phases (Custers, 2013) without any intentional effort by developers to include such biases (Caliskan et al., 2017) and could ultimately result in discriminatory decisions if their biased recommendations are accepted.

Discriminatory decisions/action can take two forms: direct and indirect. While direct discrimination involves disadvantaging someone(s) due to their membership in a protected class, indirect discrimination refers to decisions that although based on apparently neutral variables, disadvantage a protected group intentionally or unintentionally. For example, an employer refusing to hire immigrants is considered direct discrimination, and banks denying loans to residents from selected neighbourhoods, which have a much higher proportion of immigrants than average is an example of indirect discrimination.

In the context of algorithmic decision aids, direct discrimination can take place where protected characteristics (e.g., sex) are explicitly coded into an algorithm (Adams-Prassl et al., 2023) or when it learns from existing biased data sets in which such protected characteristics have been used in previous (human-made) decisions (Xing et al., 2021). However, indirect discrimination in algorithmic recommendations is related to recommendations in which protected characteristics do not play a direct role and are produced due to biased or non-representative data, and/or inadvertent modelling procedures (Žliobaitė & Custers, 2016). Similar categorisations that employ different terminologies are found in the algorithmic bias literature. For examples, Martin (2019) categorises these two types of discrimination as process mistakes and category mistakes, while Silva and Kenney (2019) term them algorithmic focus bias, and algorithmic processing bias, respectively. Given the significant differences in the nature of these two types of discrimination as well as the methods to identify and combat them (Cossette-Lefebvre & Maclure, 2022; Žliobaite, 2015), in this study, we distinguish these two types of discrimination and examine how transparency approaches can be (in)effective in addressing them.

2.2. Algorithmic transparency

Many algorithmic systems are constructed as black boxes, which hide their internal logic from users. As such, users often have little understanding of how and why these systems make certain decisions or produce specific recommendations. The transparency of these systems is important to minimise harm, improve trust, and support legal compliance (Jobin et al., 2019; Kordzadeh & Ghasemaghaei, 2022). Below we discuss two types of transparency that can play an important role in fulfilling these goals.

2.2.1. Feature importance transparency

The extant literature provides several methods of model explanations (for a review, see Guidotti et al., 2018; Linardatos et al., 2020). While there is no one best approach to explanations (Arya et al., 2020), a few methods have gained popularity. FI is one such popular method due to its advantages over other approaches (e.g., decision trees, decision rules) in terms of interpretability, scalability, and capability to reveal the relative importance of various features, etc (Huysmans et al., 2011; Lipton, 2016). FI refers to identifying variables that are used in an algorithm and revealing how these variables are used and weighted in the algorithm. This approach can help pinpoint areas of bias that may be otherwise concealed (Kirkpatrick, 2016). As such, increasing the transparency of algorithms via the FI approach can impact users' perceptions of their fairness (Grgić-Hlača et al., 2018). Specifically, users can assess each feature used in generating recommendations in terms of different criteria such as relevance, reliability, and voluntariness (i.e., relating to decision subjects' own will, not uncontrollable circumstances) – and consequently form their opinions in light of that knowledge regarding the algorithm's fairness (Albach & Wright, 2021; Grgić-Hlača et al., 2018; Scurich & Monahan, 2016). By enabling users to learn about the factors that drive the generation of algorithmic recommendations, FI fulfils users' requirement to know why a system provides a specific recommendation, a goal that Schwartz et al. (2022) suggest is more important for users than learning how an algorithmic system works. Overall, if users cannot "see" the features an algorithm employs, it is difficult if not impossible to assess the potential biases that its outputs could cause (Gal et al., 2022).

2.2.2. Demographic-based transparency

The extant body of literature on algorithmic decision aids has shown that users are generally against using demographic features by algorithms. Thus, users' perceived fairness of an algorithm is significantly hampered when they realise a demographic characteristic is used to generate the algorithm outputs (Nyarko et al., 2021; Van Berkel et al., 2021; Woodruff et al.,

2018). This realisation can take place when receiving such transparency information via pictorial and/or textual exhibits about features used by algorithms.

Another type of demographic-based transparency focuses on distributions of data sets and how they are different/similar across different demographic groups. For instance, Ebrahimi and Hassanein (2021) propose the concept of demographic transparency graphs, which provide users with information about the proportion of each demographic group (e.g., men and women) in its pertinent demographic category (e.g., sex) in the original data set and the algorithmically recommended sample to receive a benefit. They find that such transparency information decreases users' acceptance of discriminatory algorithmic recommendations. A related yet different approach involves providing information on how a data set is distributed in relation to the decision variable. For instance, Dodge et al. (2019) contend that providing summary data about the proportion of cases from a particular demographic group (e.g., Caucasian, African American) who share a label (e.g., recidivism) in a training data set impacts users' fairness perceptions.

Overall, whereas the literature includes various approaches to demographic-based transparency, the findings are generally in agreement indicating that users are against using features and methods that lead to discriminatory outputs by algorithms. Furthermore, findings depict that users' perceived fairness of and trust in algorithms are even more important than their actual performance (M. K. Lee, 2018). This is why the literature distinguishes the technical and social fairness perspectives in relation to algorithmic fairness (Dolata et al., 2022). In the next section, we discuss in greater detail the notion of fairness as well as individuals' perceived fairness of events and decisions and their consequences in terms of individuals' behaviour.

2.3. Fairness and ethical behaviour

2.3.1. Fairness

The concept of fairness focuses on justice and the equal treatment of individuals and groups. The presence of bias has negative impacts on individuals' perceptions of fairness (Leventhal, 1980) and its absence increases individuals' fairness perceptions (e.g., R. Wang et al., 2020). Fairness perceptions are often influenced by different sets of rules (Colquitt & Zipay, 2015). For example, distributive fairness focuses on the fairness of resource distribution (Greenberg, 1987) and focuses on evaluations of the fairness of outcomes (Adams, 1965). In contrast, procedural fairness emphasises the justice of the resource-distribution decision-making process and emphasises the fairness of the process by which outcomes are produced (Lind & Tyler, 1988). Specifically,

procedural fairness considers a decision as fair when the decision-making process meets certain criteria such as accuracy, lack of bias, correctability, and ethicality (Leventhal, 1980). The extant literature has shown that perceptions regarding these two types of fairness are independent, yet highly correlated (for meta-analytic reviews on this topic, see Cohen-Charash & Spector, 2001; Colquitt et al., 2001).

In organisational settings, such fairness perceptions have been shown to subsequently, impact employees' behavioural intentions, actions, trust, and satisfaction through their effects on attitudes (Cohen-Charash & Spector, 2001). For instance, distributive fairness significantly affects attitudes towards specific outcomes in question, while procedural fairness impacts attitudes towards institutions and authorities (Folger & Konovsky, 1989; Sweeney & McFarlin, 1993). Indeed, fairness perceptions regarding situations can impact perceptions of social entities, which can in turn impact employees' cognitive, affective, and behavioural responses (Cropanzano et al., 2001).

In the context of algorithmic decision aids, it is argued that people generally see algorithms as fair (M. K. Lee, 2018; Sundar & Nass, 2001). However, the presence of biases in algorithmic recommendations means that they distribute benefits (or burdens) unequally among individuals from different groups, thereby bringing about discrimination (Giovannola & Tiribelli, 2022), which negatively impacts users' fairness perceptions (Kordzadeh & Ghasemaghahi, 2022; Saxena et al., 2019). Indeed, discriminatory cases are typically seen as unfair, which cause human reactions such as ethical outrage (Bigman et al., 2023).

2.3.2. Ethical behaviour

Ethics and ethical behaviour are the subjects of numerous studies in psychology, sociology, business, and information systems. One of the well-cited theories across various fields is Rest's (1986) theory of ethical decision-making, where individuals must first recognise the ethical issue (i.e., moral awareness), form an ethical judgment (i.e., consider and evaluate different courses of action based on their morality), and prioritise ethical concerns over other concerns (i.e., establish ethical intent) before eventually acting on their ethical intentions. Rest's theory has been extensively used in various contexts to explain individuals' engaging in ethical behaviour or refraining from committing an unethical behaviour (Craft, 2013).

Rest's (1986) first and pivotal step in ethical decision-making is conceptualised in the literature as moral awareness, which is defined as "the person's determination that a situation should encompass moral content that deserves consideration from a moral point of view" (Al Halbusi et al., 2021, p. 607). Moral awareness is a critical foundation for the subsequent stages of ethical decision-making and

action (i.e., ethical judgment, ethical intention, and ethical behaviour) (T. M. Jones, 1991). For instance, moral awareness has been shown to precede individuals' reporting of peers' dishonest academic behaviour (Rangkuti et al., 2022), and engagement in whistleblowing behaviour (Manesh et al., 2024). In addition, moral awareness has been shown to negatively impact individuals' decisions to engage in vindictive negative word of mouth (He & Harris, 2014), unethical pro-organisational behaviour (Nosrati et al., 2023), and deviance from an organisation's directions (Gok et al., 2017). However, it is important to note that many decisions tend to be ethically ambiguous such that decision-makers do not always recognise that they are dealing with an ethical dilemma (Trevino & Brown, 2004).

In the context of using algorithmic systems, although a recent survey uncovered that most practitioner users are generally aware of ethical principles with respect to these systems (Pant et al., 2023), empirical studies report that users are likely to adopt biased algorithmic outputs thereby turning them into biased decisions (Ebrahimi & Hassanein, 2019). Rest's theory holds that there are moral standards with which behaviours can be compared (Barclay & Markel, 2009). Indeed, social consensus around the (un)ethicity of a decision or behaviour is an important factor in enhancing individuals' moral awareness and increasing the probability of their engagement in an ethical behaviour (Bateman et al., 2013; T. M. Jones, 1991; D. Wang et al., 2021). However, achieving such consensus could be difficult in opaque and/or uncertain situations. Therefore, researchers and practitioners have turned to algorithmic transparency, without which determining the appropriateness of deploying algorithms in different circumstances is difficult, if not impossible (Gal et al., 2022).

In light of this discussion, we employ Rest's theory to examine the impact of increasing the transparency of algorithmic recommendations, which can reveal their biases, if any, on users' moral awareness and ultimately their intention to engage in an ethical behaviour (i.e., intention to not approve biased recommendations).

2.4. Trust in algorithmic decision aids

Trust in algorithmic decision aids refers to a user's confidence in and willingness to act based on the recommendations and decisions of the system (Drozdal et al., 2020). The extant literature shows somewhat conflicting results regarding users' trust in algorithmic systems. While some studies have reported users' preference for algorithmic over human judgment (Bogert et al., 2021; Logg et al., 2019; Thurman et al., 2019), other research suggests that humans have shown numerous instances of low

trust in algorithmic predictions, resulting in humans overriding the advice of these systems, a phenomenon called "algorithmic aversion" (Dietvorst et al., 2015). This is particularly true when users realise that an algorithm has made a mistake, which often causes them to lose trust in the algorithm (Dietvorst et al., 2015). Notwithstanding the state of users' trust in algorithmic decision aids, the important role of users' trust in the success of algorithmic systems has been widely acknowledged (Abbasi et al., 2016; Felzmann et al., 2020; Smuha, 2019).

To explain trust relationships between users and algorithmic systems, two research streams have emerged in the literature. The first stream draws on the concept of interpersonal trust in order to conceptualise trust in human-like IT systems (Lankton et al., 2015). This adaptation of interpersonal trust has its roots in theories of social response to computing (Thatcher et al., 2013). The second stream, which focuses on trust in automated systems, takes a different stance and develops three bases for users' trusting beliefs – performance, process, and purpose (J. D. Lee & See, 2004). While performance focuses on competency or expertise demonstrated by the system's ability to achieve the user's goals, purpose focuses on the extent to which the system is used within the realm of the designer's intent, and process emphasises the extent to which the system's algorithms are appropriate for the situation and able to achieve the user's goals (J. D. Lee & See, 2004). Therefore, while performance illustrates *what* the algorithmic system does, purpose refers to *why* the system was developed, and process focuses on *how* the system perform its tasks. This implies that humans tend to trust automation if its internal workings (i.e., algorithms) are transparent and explainable (Samek et al., 2019), can be understood, and are able to achieve the intended goals (Dizaji & Hu, 2021).

In line with process-based trusting beliefs, which focus on the concept of understandability in human – automation relationships (Gebru et al., 2022; J. D. Lee & See, 2004), the extant body of literature on algorithmic decision aids has widely acknowledged the importance of transparency in determining users' trust in a system. For instance, Thiebes et al. (2021) propose explicability, which focuses on systems transparency and accountability, as one of the five foundation principles of trustworthy artificial intelligence. Several other studies (e.g., Krasnova et al., 2014; Springer & Whittaker, 2019) argue that transparency of algorithms is essential for trusting them as system outputs that are not well-explained are less acceptable and decrease trust in algorithmic decision aids (e.g., Grimmelikhuijsen, 2023; Kaplan et al., 2023; Miller, 2019; Rudin, 2019).

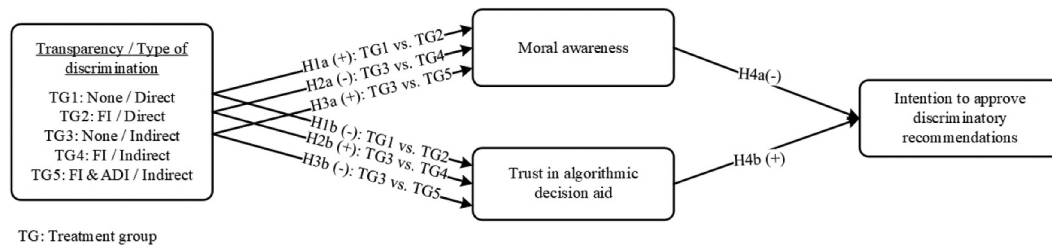


Figure 1. Research Model.

3. Research model and hypotheses

To answer our research question, we examine the extent to which providing users with transparency regarding potentially² biased algorithmic recommendations impact their approval of such recommendations. To this end, we focus on the impacts of increasing transparency through providing FI on users' cognitions and behavioural intention in the two contexts of direct and indirect discriminatory recommendations. Additionally, we study whether providing ADI as an additional level of transparency that sheds light on the distribution of important features against sensitive characteristics (e.g., sex) influences users' cognitions and behavioural intentions. The following sections present and delineate our hypotheses, as depicted in Figure 1.

3.1. Feature importance when dealing with biased algorithmic recommendations

FI is a simple but effective tool to provide explanations regarding the features used by an algorithm and their relative contributions to the generation of its outputs (Guidotti et al., 2018). Various providers have begun to incorporate FI as a means to increase the transparency of their algorithmic systems (e.g., SAP and Oracle predictive analytics). Previous research has shown that the transparency provided by FI can enable users to better understand and simulate algorithms' predictions (Poursabzi-Sangdeh et al., 2021). Similarly, we suggest that FI can help users detect biased algorithmic recommendations when they include direct discrimination. As an example, consider the case of an organisation that uses a machine learning model to decide whom to promote. It is possible that the model, based on the preference of (previous) managers found in historical data (e.g., previous promotions) learns that marital status is an important variable when making such decisions (Martin, 2019). Including this variable in the algorithm (when/if the model is trained using historical data that prefer single employees) would be discriminatory against married employees and is an instance of direct discrimination. If this system does not provide users with information about what variables drove its recommendation of

employee(s) to be promoted, then chances are low that users can identify the bias in the recommendation. By providing FI, users will be equipped to figure out the presence of the bias, which will in turn impact their perceptions of the fairness of the recommendation generation process. Similarly prior research has shown that users perceive an algorithm as unfair when they find demographic variables among features it uses to produce outcomes (Nyarko et al., 2021; Van Berkel et al., 2021). Users' realisation of the lack of procedural fairness then triggers their moral awareness as they are more capable of and likely to reason from common moral principles (Konow, 2009). Indeed, people's reaction to unfairness have roots in our evolutionary history and can elicit moral outrage (Bigman et al., 2023; Spring et al., 2018).

An individual's observations of procedural (un) fairness also impact their perceptions of the pertinent social entity as well as their cognitive, affective, and behavioural responses (Cropanzano et al., 2001). Specifically, procedural fairness has been observed to strongly predict trust in organisations (Cohen-Charash & Spector, 2001; Colquitt et al., 2001). Individuals possess cognitive schemas that represent just and unjust behaviours, the violation of which is considered a signal of unfair conduct (Lind, 2001). Given that a user's belief regarding the process of how an algorithmic system performs its tasks is one of main foundation of users' trust in it (J. D. Lee & See, 2004), observations of unfair conduct by the system can negatively affect users' trust (D. A. Jones & Martens, 2009).

Therefore, if/when a user is provided with FI sees signs of procedural unfairness in an algorithm's analysis (i.e., finding a sensitive variable such as marital status included in the black box model an algorithmic system has employed to generate recommendations), the user is likely to be concerned about the system's reliable functionality in the role of a decision aid. Indeed, previous research has shown that individuals have concerns when demographic characteristics are utilised by algorithms (Woodruff et al., 2018). Given that a system's functionality reduces task-related risks and, therefore, improves users' trust in it (Lankton et al., 2014), identifying such biases is likely to hamper users' trust in the system's competence and integrity.

Furthermore, there is a common belief that, unlike humans, machines are unbiased (M. K. Lee, 2018; Sundar & Nass, 2001), which gives rise to the expectation that algorithms do not consider sensitive variables in their analysis. However, a violation of this expectation can be considered an instance of a service failure, which negatively impacts users' trust (Goles et al., 2009; Malhotra et al., 2017). Thus, we posit the following hypothesis:

H1: *When receiving a biased algorithmic recommendation that includes direct discrimination, users who are provided with FI will have (a) higher moral awareness, and (b) lower trust in the system compared with those who do not receive FI.*

FI, however, is not as likely to help when biased algorithmic recommendations include indirect discrimination as opposed to direct discrimination. In such a situation, biased recommendations are generated due to reasons other than the inclusion of sensitive variables in generating an outcome. Non-sensitive variables acting as proxies for sensitive variables is one such reason (Pedreschi et al., 2008). In such cases, FI would reflect a set of neutral variables. We suggest that this leads to lowering users' moral awareness as even if users have reservations about using a model for decision-making, after seeing a list of non-sensitive variables (e.g., tenure in the company), they are likely to feel at ease about issues related to fairness. This is because users might have concerns regarding algorithms' fairness on account of not understanding their internal processes and/or feeling that they do not rely on the appropriate information in the appropriate manner (Poursabzi-Sangdeh et al., 2021). However, these concerns, can be ameliorated by FI providing information about the apparently neutral nature of the basis for an algorithm's recommendations. This is in line with previous empirical research, which has shown that after receiving evidence of absence of biases (i.e., similar error rates across different demographic groups), users' perceptions of the fairness of systems are strongly increased (R. Wang et al., 2020).

Increasing the transparency of an algorithm by providing FI, which shows a seemingly neutral set of variables, can also impact users' trusting beliefs towards the tool. Increasing users' trust in the recommendations of intelligent systems' by providing explanations has a long history in the IS literature. By virtue of shedding light on the performance of a system, explanations are influential in improving users' trust in the system and their acceptance of it (Ribeiro et al., 2016; Venkatesh et al., 2016; W. Wang & Benbasat, 2016). When users of a black box system are presented with FI, they receive information that reduces some of the uncertainty surrounding how the system has

generated its recommendation. Users tend to feel more at ease when they better understand a model, even if this understanding does not appear to serve a clear purpose (Lipton, 2016). Previous research has shown that the manipulation of FI information for a biased model to exclude sensitive variable (e.g., age and race) and include desired features (e.g., prior jail incarcerations) increases users' trust in the model (Lakkaraju & Bastani, 2020). A similar situation is likely to take place in the case of biased outputs in the form of indirect discrimination as information conveyed through FI does not include any directly unethical or sensitive aspects, and therefore is likely to enhance users' trust in the system. Thus, in line with previous empirical research which suggests the positive impact of algorithmic transparency on users' trust (e.g., Shin et al., 2020), we posit:

H2: *When receiving a biased algorithmic recommendation that includes indirect discrimination, users who are provided with FI will have (a) lower moral awareness, and (b) higher trust in the system compared with those who do not receive FI.*

3.2. Feature importance and aggregated demographic information

As was discussed in Section 3.1, increasing transparency by using FI when an algorithmic recommendation includes indirect discrimination is not only unhelpful but can also backfire as it falsely reassures users about the ethicality of recommendations. Because algorithms and their recommendations should be understood in "the wider context of their sociotechnical assemblage" (Kitchin, 2017, p. 18), understanding algorithmic bias requires looking beyond the data and into the larger sociotechnical systems that encompass the data (Adomavicius & Yang, 2022). As such, FI does not always provide a complete picture of the circumstances surrounding the generation of a set of recommendations (Balakrishnan et al., 2022) and thus might not reveal the potential of bias contaminating these recommendations.

Therefore, we suggest accompanying FI by ADI that provides certain information about the distribution of each of the important features included in the analysis and, therefore, FI for each demographic group (e.g., women and men). Thus, ADI reveals to users if a feature included in FI has significantly different values for different demographic groups. As an example, consider the case of using an algorithmic system for recommendations regarding a monetary bonus decision in an organisation. The system, along with its recommendations, provides FI plus ADI, which reveals that the average number of annual leave days taken makes a contribution of 31% to the

recommendation and that the distribution of this feature is significantly different for men and women, such that men, on average, have significantly fewer annual leave days. Such a situation can take place if this variable includes both vacation leave and parental leave as women are more likely to take parental leave (Ebrahimi & Matt, 2023). Upon examination of this additional information (i.e., FI plus ADI), users are then likely to see that women have not been offered an equal opportunity to receive the monetary bonus. This observation can then help users discover the (potential) discrimination related to the decision at hand. In addition, users will be empowered to see what feature(s) included in the analysis treat(s) a disadvantaged demographic group negatively and therefore, will be equipped with more information to judge the (un)fairness of algorithmic recommendations.

Therefore, providing ADI along with FI will enable users see the lack of equality of opportunity for individuals from different demographic groups. Equality of opportunity, a well-known fairness metric, measures whether individuals who should qualify for an opportunity have the same likelihood of being deemed qualified by an algorithm regardless of their demographic group (Morse et al., 2022) and can be understood in terms of the criteria that are permissible and impermissible in determining qualified individuals for the opportunity in question (Elford, 2023). The observation of a lack of equality of opportunity is then likely to increase the likelihood of users noticing the possibility of indirect discrimination in a recommendation. More specifically, if ADI reveals that significant differences exist between the distribution(s) of one or several important feature/s in the disadvantaged class (e.g., women) and the corresponding distribution(s) in the other class(es), then users are more likely to doubt the fairness of the recommendation generation process and become morally alarmed (Grgić-Hlača et al., 2018).

Selection of inappropriate variables, which are not considered illegal or unacceptable but are highly correlated with sensitive features and, thus, can act as their proxies is still an instance of model bias (Robert et al., 2020) which signals poor modelling procedures (Žliobaitė & Custers, 2016). In addition, such observations of unfair algorithmic performance can have ramifications for users' trust. In fact, previous research has shown that users' perceived fairness of an algorithm has a direct impact on their trust in the system (e.g., Shin, 2020) and lack of fairness in algorithmic recommendations lowers users' trust in the system (Zhou et al., 2021). Therefore, users' finding discrepancies in the distribution of features used by an algorithm against sensitive variables can negatively impact their perceptions of the system's capability to complete the task in a reliable manner. Thus, we posit:

H3: *When receiving a biased algorithmic recommendation that includes indirect discrimination, users who are provided with FI and ADI will have (a) higher moral awareness, and (b) lower trust in the system compared with those who do not receive FI.*

3.3. Moral awareness, trust, and intention to accept biased recommendations

Moral behaviour has been portrayed as a systematic framework which entails making ethical assessments in unclear and morally charged situations (Ferrell et al., 1989; Rest, 1986). According to Rest's (1986) theory of ethical decision-making, the first phase in making an ethical decision for an individual is becoming morally aware. When individuals recognise the ethical aspect of a decision/situation, they then consider and assess various courses of action in terms of their ethicality. Subsequently, these assessments impact individuals' ethical intentions and actions. Empirical evidence for this systematic framework has been observed in numerous contexts such as individuals' purchases (Husser et al., 2019), merchandise return (Chang & Yang, 2022), online fraud (Harrison et al., 2020), and various information technology-related behaviours (Ulman et al., 2021). In the context of algorithmic decision aids, when users learn about potential biases in algorithmic outputs, they tend to take steps to ameliorate the biases (Ebrahimi & Hassanein, 2021; Luong et al., 2021; Rhue, 2019). Consequently, we expect users who become aware of the bias in algorithmic recommendations to be more likely to feel accountable regarding the potential negative consequences of approving the recommendations and, thus, be less likely to approve them as doing so would convert the biased recommendations into biased decisions.

H4(a): *Users' moral awareness is negatively associated with their intention to approve discriminatory recommendations.*

The extant IS literature has long examined the impact of trust in an information source and users' adoption intentions (Ebrahimi et al., 2022; Gefen et al., 2003). Indeed, trust as a confident expectation in "situations of risk that one's vulnerabilities will not be exploited" (Corritore et al., 2003, p. 740), plays a pivotal role in the assessment and selection of information sources (Hertzum et al., 2002). Similarly, when interacting with intelligent decision aids, trust, is an important decisive factor in users' intention to accept their recommendations (Benlian et al., 2012; Komiak & Benbasat, 2006). Users' trust in an algorithmic system involves the confidence and the expectation that the system will fulfil its promises, meet users' needs in

Table 1. Treatment Groups.³

Treatment group (TG)	Type of discrimination	Transparency	No. of participants
1	Direct	None	53
2	Direct	FI	51
3	Indirect	None	48
4	Indirect	FI	47
5	Indirect	FI + ADI	51

a positive manner, and safeguards users' interests in the case of risks or failures (Lv et al., 2022; Park, 2020). Therefore, the extant literature reports a significant effect of users' trust in algorithmic systems on their attitudes, experiences, and intentions (for a review, see Yang & Wibowo, 2022). Thus, we propose the following hypothesis:

H4(b): *Users' trust in an algorithmic system is positively associated with their intention to approve discriminatory recommendations.*

4. Research method

To test the hypothesized relationships, we conducted a between-subjects online experiment. Participants were randomly assigned to one of five experimental groups. These groups varied in terms of the type of discrimination in the recommendations and in the presence and type of transparency offered by the experimental algorithmic system (see Table 1).

4.1. Experimental task

After receiving research ethics approval from the Human Participants Review Committee of the first author's university, participants were invited to interact with an online (fictitious) algorithmic system called "Smart HR-Tech" that was designed to provide fictitious yet realistic HR-related recommendations for selecting employees who were to be provided with the opportunity to attend a leadership training programme. To increase participants' motivation to take the task seriously, they were told that the "training programme is a critical criterion for promotions to senior level management within the organisation".

Not being offered training, particularly during the earlier stages of individuals' careers, due to biased algorithmic recommendations can have detrimental impacts on the subsequent years of their career (Mikalef et al., 2022).

During the experimental task, participants first received an introductory script that explained the task as well as the functionalities of Smart HR-Tech. Subsequently, they received a list of the variables used in the data set along with their definitions. This was followed by participants' receipt of a list of recommended employees that Smart HR-Tech suggested should be provided with the opportunity of attending the training programme. This list was displayed in a manner that participants could toggle between viewing all employees and recommended employees. Participants could also examine the data set using various methods such as sorting and/or filtering based on various features (e.g., education) (See Figure 3). Participants were given an unlimited amount of time to examine the entire data set of all employees as well as the subset of the data set containing the tool's recommendations and verify the suitability of the recommendations. Subsequently, participants in different treatment conditions were provided with pertinent transparency information (i.e., FI, FI + ADI), if any. When participants were ready to exit the experimental tool, they were directed to the online questionnaire. Figure 2 illustrates the experimental procedure of this study.

4.2. Treatment conditions and sample collection

Our experimental algorithmic system generated recommendations based on a multiple linear regression of several variables. For participants in the treatment group (TG) 1 and TG 2, who received recommendations with *direct* discrimination, the recommendations were based on the seven variables of education, sex, annual performance rating, years of employment at the company, number of companies worked at, number of job grade promotions, and business travel frequency. Given that the variable sex played a role in generating the tool's output, such that male employees were given an advantage, the

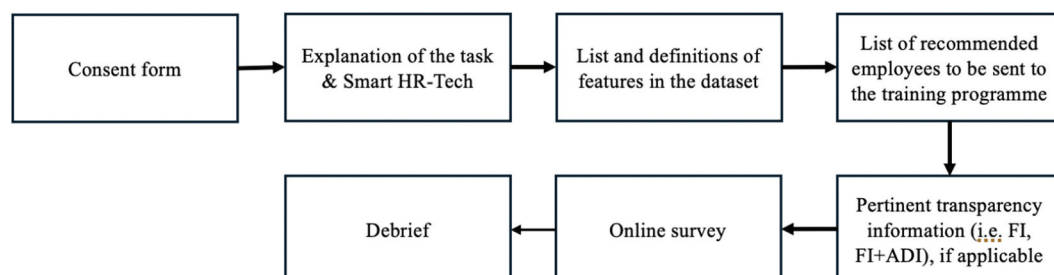


Figure 2. Experimental procedure.

Select the Employee Data to View:

☐ View Entire Data Set of All Employees ☒ View Recommended Employees

Additional Data Filters

Show entries

Search:

Business travel frequency:

☐ 1
☐ 2
☐ 3

Education:

☐ 1
☐ 2
☐ 3
☐ 4
☐ 5

Selected /Not Selected	ID	Business travel frequency	Education	Birthdate	Sex	Performance rating	Number of companies worked at	Percent salary increase in the last 5 years	Number of job grade promotions in the last 5 years	Years of active employment at the company
1	62	3	4	1982-06-13	M	5	4	21	5	9
1	74	3	5	1994-07-21	M	3	1	22	4	9
1	112	2	3	1985-01-25	M	5	4	11	4	6
1	163	2	4	1990-01-21	M	5	4	18	4	7
1	215	2	4	1979-12-01	M	5	7	11	4	12
1	243	2	4	2002-11-30	M	5	1	5	4	1
1	267	2	3	1995-12-13	M	4	1	12	5	8
1	339	2	4	1982-08-18	M	2	6	13	2	11
1	381	2	3	1978-06-27	W	4	7	18	2	5

Figure 3. A portion of the data set of (all/recommended) employees in the systems.

recommendations included discrimination against female employees. Specifically, out of 300 (150 male and 150 female) eligible employees, 30 (22 male and 8 female) employees were selected for the training programme.

For participants in TG 3, TG 4, and TG 5, who received recommendations with *indirect* discrimination, the recommendations were based on a similar set of variables to those in TG 1 and TG 2 except for the inclusion of the variable sex. To generate recommendations with indirect discrimination, the two variables of active years of employment at the company and number of job grade promotions were adjusted to be, on average, lower for female employees than their male counterparts. These two variables are among those that are known to disadvantage female employees as the former can be significantly higher for women due to the higher likelihood of them going on parental leave and the latter can disadvantage women due to the well-documented “glass ceiling”, which causes invisible obstacles to women’s promotion to top positions (Cook & Glass, 2014; Cotter et al., 2001). As a result of the inclusion of these two variables in the analysis, the recommendation generated by our experimental tool was discriminatory against female employees. The number and proportion of male and female employees in both the pool of all eligible employees and those recommended for the training opportunity were identical to the ones for TG 1 and TG 2.

After receiving the recommendations regarding which employees should be provided with this opportunity, participants in TG 2 and TG 4 received FI (see Figures 4 and 5, respectively), which illustrated a list and the percentage contribution of the different features used by the system to generate the

recommendation. Participants in TG 5 received the same FI information as those in TG 4, which was accompanied by ADI (i.e., a graph that visually depicted the average of different features for men vs. women) (see Figure 6).

We recruited 250 participants via Qualtrics-a market research firm. This sample size assured adequate statistical power of 80% to detect a medium effect size of 0.25 (Cohen, 1988). Participants were all middle managers employed in North America, who had more than one employee reporting to them. Among the participants, 30% were women and 70% were men. On average, participants were aged 43.5 years. The distribution of participants’ highest education level was 6% high school, 12% college diploma, 48% bachelor’s degree, 30% master’s degree, and 4% doctorate degree. A breakdown of participants across various treatment groups is depicted in Table 1.

4.3. Measures

All constructs were measured using established multi-item scales (listed in Appendix A). Moral awareness was measured using May et al. (2014) four-item scale. The three-item scales for trust and intention to approve recommendations were adapted from Cyr et al. (2009) and Venkatesh et al. (2003), respectively. In this study, the impacts of participants’ gender and education level on their intention to accept discriminatory recommendations were controlled for. Furthermore, to account for the potential effect of participants’ tendency for socially desirable behaviour, we controlled for the impact of their impression management, defined as the propensity to “consciously over-report their performance of a wide variety of desirable behaviours and under-report undesirable

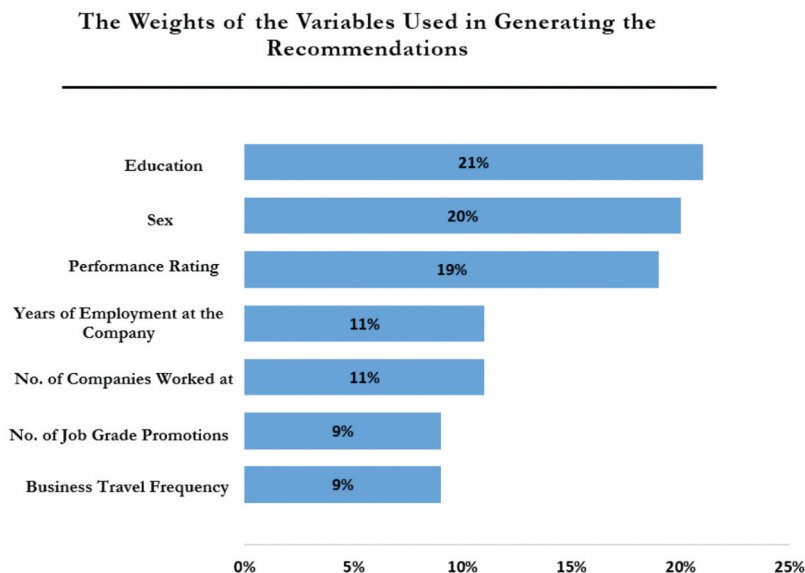


Figure 4. FI (provided to participants in TG2).

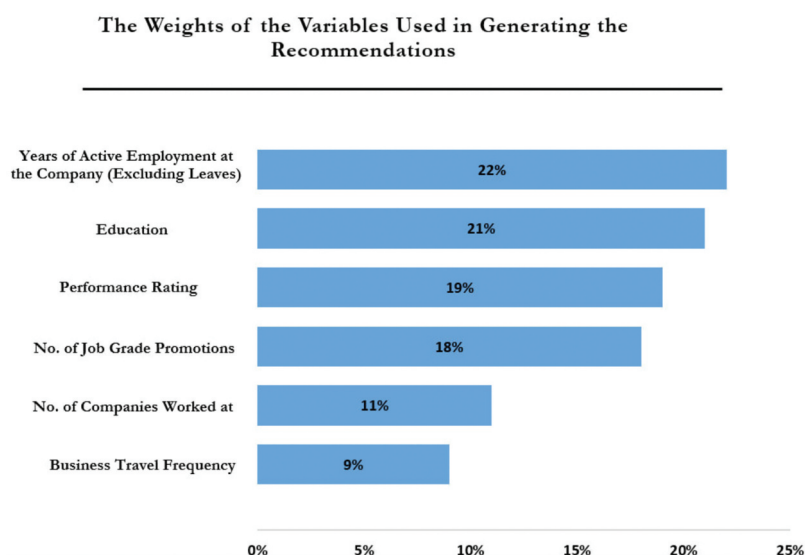


Figure 5. FI (provided to participants in TG4 and TG5).

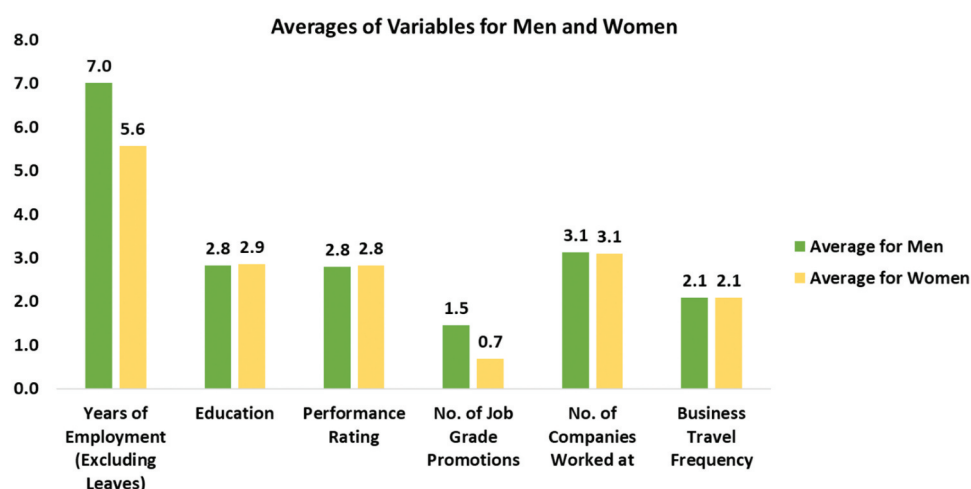


Figure 6. ADI (provided to participants in TG 5).

behaviour” (Paulhus, 1984, p. 4). Impression management has been shown to have a positive impact on self-reported ethical conduct (e.g., Dalton & Ortengren, 2011; Schoderbek & Deshpande, 1996). To that end, we used a six-item scale developed by Paulhus (1984). Following Paulhus’ approach, we first reverse coded the negatively worded items. Subsequently, we added one point for each extreme response (i.e., 6 or 7 on the Likert scale). The approach yields the minimum score in impression management (i.e., 1) when a participant does not provide an extreme response for any of the questions and the maximum score (i.e., 7) when a participant provides extreme responses for all questions.

Furthermore, we controlled for participants’ sensitivity to moral issues by measuring their moral attentiveness. Moral attentiveness is defined as “the extent to which an individual chronically perceives and considers morality and moral elements in his or her experiences” (Reynolds, 2008, p. 1028) and has been empirically shown to increase individuals’ moral engagement and behaviour (Al Halbusi et al., 2021; Whitaker & Godwin, 2013). Reynolds (2008) further suggests that moral attentiveness comprises two main components: perceptual and reflective. While perceptual moral attentiveness has been empirically shown to predict the recall and reporting of morality-related behaviours, reflective moral attentiveness is associated with moral behaviour (Reynolds, 2008; Sturm, 2017). As such, in this study, we measured and controlled for the impact of participants’ reflective moral attentiveness using Reynolds’s (2008) five-item scale.

4.4. Manipulation checks

We conducted three manipulation checks to assess the validity of our experimental treatments. First, we analysed responses by participants in TG 1 and TG 2 on two questions about their perceptions of the FI transparency of Smart HR-Tech (see Appendix A). The result of our independent samples t-test revealed that the manipulation was successful ($\text{Mean}_{\text{TG1}} = 4.79$ versus $\text{Mean}_{\text{TG2}} = 5.41$, $p < 0.01$). Similarly, participants in TG 4 found the Smart HR-Tech’s FI transparency to be higher than those in TG 3 ($\text{Mean}_{\text{TG3}} = 4.92$ versus $\text{Mean}_{\text{TG4}} = 5.75$, $p < 0.001$). Finally, we analysed whether participants in TG 5 perceived Smart HR-Tech to have a higher FI + ADI than those in TG 3. The results confirmed our expectation ($\text{Mean}_{\text{TG3}} = 4.98$ versus $\text{Mean}_{\text{TG5}} = 5.75$, $p < 0.005$).

4.5. Preliminary analyses

Prior to testing our hypotheses, preliminary data analyses were performed to test for reliability, normality, convergent and discriminant validity, and common

method bias. First, we assessed the reliability of all constructs using Cronbach’s alpha and observed that all constructs meet the acceptable reliability threshold of 0.7 (see Appendix B, Table B1) (Hair et al., 2010). Furthermore, we examined the normality of all variables and found no violations.

Confirmatory factor analyses of the intention to approve recommendations, trust in the system, and moral awareness variables using AMOS 28.0 indicated a good fit and supported convergent validity for a three-factor model ($\text{CMIN}/\text{df} = 1.36$, $\text{CFI} = 0.99$, $\text{NFI} = 0.98$, $\text{TLI} = 0.99$, $\text{RMSEA} = 0.04$), all item-factor loadings ≥ 0.74 ($p < 0.001$) (see Appendix B, Table B2) (Anderson & Gerbing, 1988). Discriminant validity was assessed using Fornell and Larcker’s (1981) criterion. Since the square root of the average variance extracted (AVE) for each construct is greater than the correlation between the construct in question and other constructs, we can conclude that our constructs indicate an acceptable level of discriminant validity.

In addition, we performed two tests to address the potential concern for common method bias (CMB). First, we tested for multicollinearity among our three constructs of interest (i.e., moral awareness, trust, and intention to approve recommendations). The variable inflation factor (VIF) values were well below the threshold of 10 (Petter et al., 2007), with the highest VIF being 1.20. Second, following Lindell and Whitney (2001), we employed the marker variable technique by using a theoretically unrelated construct to adjust the correlations among the main constructs. We used participants’ coffee drinking habits as the marker variable and observed the average correlation between the three main constructs in our study and the marker variable to be 0.002 (average p -value = 0.34). We found that after adding the marker variable, all correlations remain significant. The results of the multicollinearity test and the marker variable analysis suggest that common method bias is not a concern in this study.

5. Results

5.1. Hypotheses assessment

We conducted linear regression using IBM SPSS Statistics (version 28) to test our hypotheses. H1a and H1b focus on the influence of the presence of FI on users’ moral awareness and trust, respectively, when algorithmic bias is in the form of direct discrimination. Therefore, for H1a and H1b, transparency in the direct discrimination context was the between-subjects factor (no transparency for TG 1—the control group versus FI transparency for TG 2—the treatment group). As indicated on Table 2, for H1a, the regression analysis revealed a significant impact in relation to providing FI on users’ moral awareness. Similarly,

Table 2. Linear regression to test H1, H2, and H3.

	H1a	H1b	H2a	H2b	H3a	H3b
Constant	3.46**** (.16)	5.44**** (.15)	3.62**** (.18)	5.52**** (.13)	3.62**** (.19)	5.52**** (.15)
Unstandardised B	.71*** (.22)	-.57 ** (.21)	-.47 ^{ns} (.25)	.32 ^{ns} (.19)	.70** (.27)	-.38 ^{ns} (.22)
Standardised β	.30***	-.25**	-.19 ^{ns}	.17 ^{ns}	.26**	-.18 ^{ns}
Adjusted R ²	.08	.05	.03	.02	.06	.02
F Change	10.11***	6.98**	3.43 ^{ns}	2.85 ^{ns}	6.85**	3.09 ^{ns}

* $p < .05$, ** $p < .01$, *** $p < .005$, **** $p < .001$; two-tailed p-values.

For H1a and H1b, a dummy variable with TG1 serves as the baseline comparison between TG1 and TG2. For H2a and H2b, a dummy variable with TG3 serves as the baseline comparison between TG3 and TG4. For H3a and H3b, a dummy variable with TG3 serves as the baseline comparison between TG3 and TG5. Values represent regression coefficients, with standard values in parentheses.

for H1b, the results indicate a statistically significant impact on users' trust in the system as a result of providing FI. Specifically, increasing the transparency by providing FI (i.e., TG2) increased users' moral awareness by 0.71 and decreased their trust by 0.57 compared to the respective values in the control group (i.e., TG1), thereby supporting H1a and H1b, respectively. Given that our measurements of moral awareness and trust were based on a seven-point Likert scale, this finding indicates an increase of over 10% in users' moral awareness and a decrease of over 8% in users' trust in TG2 compared to TG1. In other words, when users, through FI, learned about direct discrimination in recommendations of Smart HR-Tech, their awareness of the moral aspect of the decision at hand increased by 10% and their trust in the system decreased by 8%.

H2a and H2b suggest differences between users' moral awareness and trust respectively, as a result of the presence of FI when algorithmic bias is in the form of indirect discrimination. Therefore, H2a and H2b had the same between-subjects factor as H1a and H1b, but in the indirect discrimination context (TG 3 and TG 4). The two regression analyses to test H2a and H2b revealed a lack of support for these two hypotheses comparing the treatment group TG4 with the control group, TG3. Specifically, the results indicate that although providing FI that encompasses a set of neutral variables decreases users' moral awareness and increases their trust in the system by 0.47 (6%) and 0.32 (4%) respectively, the extents of change are not statistically significant. This implies that when algorithmic recommendations include indirect discrimination, increasing the transparency by providing FI, which depicts a set of seemingly neutral variables due to the nature of indirect discrimination, neither increased nor decreased users' moral awareness and trust in the system.

H3a and H3b respectively claim that in the context of algorithmic recommendations contaminated by indirect discrimination, users' moral awareness and trust are affected when they receive FI accompanied by ADI, which enables them to see the potential differences between different demographic groups in terms of each feature in FI. Therefore, to test H3a and H3b, the between-subjects factor compared the case of showing no transparency (i.e., TG 3) versus providing FI and ADI (i.e., TG 5). The results (outlined in Table 2) lend support for H3a by revealing an increase in users' moral

awareness by 0.70 (equivalent to a 10% increase) as a consequence of receiving FI and ADI both. This indicates that, when FI was provided along with ADI, users had an increased awareness of the moral aspect of the decision context. However, the results fail to provide support for H3b, indicating that users' trust in the system does not change by a statistically significant extent when they see transparency-increasing information that reveals traces of indirect discrimination in systems' recommendations. In sum, according to our results, users' moral awareness increases when they see that variables used by the algorithmic decision aid have different averages for men and women and, therefore, cause biased recommendations against women. However, this observation does not impact their trust in the system.⁴

Subsequently, we conducted multiple regression to test H4a and H4b-the effects of moral awareness and trust, respectively, on participants' intention to approve discriminatory recommendations (results reported in Table 3). As outlined in the table, Model 0 includes the four control variables (i.e., gender, education, impression management, and reflective moral attentiveness) in our study. The results indicate that while the impact of gender and reflective moral attentiveness on intention to approve recommendations are non-significant, participants' education and impression management significantly impact such intentions. Then, in Model 1, we added moral awareness and trust in algorithmic decision aid, which made the impact of impression management statistically non-significant. In addition, we found that participants' intention to approve recommendations is negatively impacted by the level of their moral awareness

Table 3. Regression results with intention to approve recommendations as the dependent variable.

	Model 0	Model 1
<i>Controls</i>		
Gender (Female = 0, Male = 1)	0.10 ^{ns} (1.63)	0.06 ^{ns} (1.34)
Education	-0.14* (-2.17)	-0.10* (-2.47)
Impression management	0.18** (2.88)	-0.01 ^{ns} (-0.30)
Reflective moral attentiveness	0.00 ^{ns} (0.01)	-0.02 ^{ns} (-0.42)
<i>Predictors</i>		
Moral awareness		-0.27*** (-5.75)
Trust in algorithmic decision aid		0.60*** (12.59)
R ²	0.06	0.59
Adjusted R ²	0.05	0.58
F (df1, df2)	3.96 ** (4, 245)	57.27*** (6, 243)

* $p < .05$, ** $p < .005$, *** $p < .001$; two-tailed p-values. Values represent standardised regression coefficients, with t-values in parentheses.

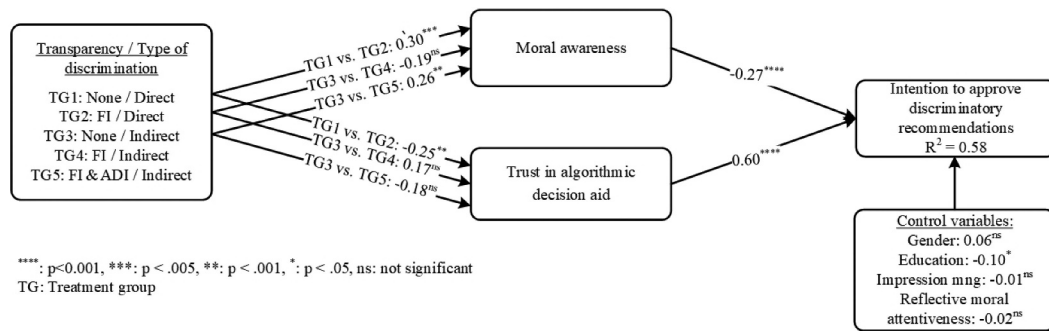


Figure 7. Results of the regression analyses.

($\beta = -0.27$, $p < 0.001$) and positively impacted by the level of their trust in the system ($\beta = 0.60$, $p < 0.001$). In other words, an increase in moral awareness decreases the adoption of discriminatory algorithmic recommendations, while trust in the system increases it, providing support for H4a and H4b. These results are also depicted in Figure 7.

5.2. Post-hoc analyses

5.2.1. The impact of transparency on intention to approve discriminatory recommendations

Since the main objective of this paper is to study whether increasing the transparency of an algorithmic system decreases users' adoption of its discriminatory recommendations, we also studied the impact of providing FI and ADI on our dependent variable, users' intention to approve recommendations. To this end, we conducted three independent sample t-tests. The first test compared users' intentions to approve recommendations that included direct discrimination for participants who received FI with those who did not receive this additional information. The results show the existence of significant differences in terms of our dependent variable between these two groups (Mean = 5.30 versus 4.54, $p < 0.001$), indicating that users who receive FI are 11% less likely to approve biased recommendations that include direct discrimination.

The second t-test focused on recommendations with indirect discrimination and compared the intention to approve them between participants who received no transparency information and those who received such information in the form FI. The results show a lack of significant differences between the two groups in terms of our dependent variable ($M = 5.30$ versus 5.63, $p > 0.05$), meaning that providing FI did not have a statistically significant positive or negative impact on participants' intention to approve these recommendations. This confirms that providing FI does not reduce the likelihood of users' approving recommendations that are contaminated by indirect discrimination.

Finally, we employed another independent sample t-test to assess the effectiveness of adding ADI to FI in terms of reducing users' adoption of indirectly discriminatory recommendations. The results provide evidence for a statistically significant difference between participants who worked with a non-transparent system and those who received FI along with ADI ($M = 5.30$ versus 4.71, $p < 0.05$), thereby indicating that the latter group was less likely to approve the biased recommendations with indirect discrimination.

5.2.2. The role of perceived fairness

The second set of post hoc analyses was conducted to verify our results and rule out potential alternative explanations for the observed impacts of increasing transparency on users' moral awareness and trust. Specifically, the objective of this set of post hoc analyses is to examine whether the changes in users' moral awareness and trust can be attributed to the changes in their fairness perceptions due to their realisation of traces of bias in the system's recommendations. To this end, we first tested whether users' perceptions of the fairness of recommendations mediated the relationships between increasing transparency and moral awareness as well as between transparency and trust.⁵ We measured perceived fairness of algorithmic output using Goodwin and Ross (1989) two-item measure for perceived fairness (see Appendix A). We then employed the PROCESS macro for SPSS (Hayes, 2018) to test the mediation role of perceived fairness using 5000 bootstrap samples. The results, detailed in Table 4, confirm that in the context of biased algorithmic recommendations that include direct discrimination, the positive impact of providing FI (i.e., TG1 vs. TG2) on users' moral awareness as well as its negative impact on their trust in the system are mediated by their perceptions of the fairness of recommendations. These results provide further evidence for our theorisation that increasing algorithmic transparency—which reveals an algorithm's use of a sensitive variable (e.g., sex)—has negative ramifications for users' fairness perceptions as it signals unfair treatments of already disadvantaged groups (e.g., women). Finding algorithmic

Table 4. The mediation role of users' perceived fairness of algorithmic recommendations that include direct discrimination.

	Coeff.	SE	LLCI	ULCI
TG1 vs. TG2* → Perceived fairness	-.52	.25	-1.02	-.01
Perceived fairness → Moral awareness	-.52	.07	-.66	-.38
TG1 vs. TG2* → Perceived fairness → Moral awareness	.27	.16	.01	.63
Perceived fairness → Trust in tool	.68	.05	.58	.78
TG1 vs. TG2* → Perceived fairness → Trust in tool	-.35	.17	-.69	-.01

*TG was coded as a categorical variable where TG1 = 0 and TG2 = 1.

recommendations as unfair, then triggers users' awareness of the morally charged nature of the situation at hand (i.e., increasing their moral awareness) and causes a decline in the user's trust in the system.

Similarly, we conducted additional analyses to study the potential mediating role of perceived fairness in the relationship between providing FI plus ADI and users' moral awareness when algorithmic recommendations include indirect discrimination (i.e., TG3 vs. TG5). The results, depicted in Table 5, indicate that the confidence interval for the indirect relationship through perceived fairness excludes zero, providing evidence for the statistical significance of the mediating role of perceived fairness. As such, in line with our expectation, providing users with FI along with ADI when algorithmic recommendations include traces of indirect discrimination empowers them to better examine the recommendation and find out that the inclusion of (a) variable(s) in the algorithm unjustly disadvantages a group of data subjects. The observation of lack of fairness in algorithmic recommendations then leads to users' becoming aware of the moral ramifications of the decision at hand.

In addition to mediation tests, we also conducted independent sample t-tests to compare fairness perceptions of participants in TG2 and TG4. Participants in both these treatment groups received FI, which differed in terms of one main aspect: the presence of the feature "sex" (see Figures 4 and 5). The results of the independent sample t-tests indicate that those in TG2 had a lower perceived fairness of recommendations than those in TG4 (i.e., Mean = 4.77 versus 5.84, $p < 0.001$), indicating that upon observing the sensitive variable sex in FI, users' perception of fairness of algorithmic recommendations deteriorates. We observed similar trends in terms of our dependent

variables in the model. Specifically, compared to participants in TG4, those in TG2 had higher moral awareness (Mean = 4.17 versus 3.15, $p < 0.001$), lower trust in the system (Mean = 4.87 versus 5.84, $p < 0.001$), and lower intention to accept recommendations (Mean = 4.54 versus 5.63, $p < 0.001$). These results, therefore, reinforce our findings in relation to the impacts of users' observation of a sensitive variable among the features used by an algorithm. Specifically, according to our findings, such an observation decreases users' perceived fairness of recommendations, triggers their moral awareness, and reduces their trust in the system. Subsequently, these changes in users' perceptions and attitudes bring about a lower likelihood of them accepting biased algorithmic recommendations.

6. Discussion

6.1. Key findings

Motivated by the question about the extent of the effectiveness of algorithmic transparency on users' adoption of discriminatory recommendation, we studied the influence of FI transparency and FI transparency augmented by ADI transparency in the two contexts of direct and indirect discrimination. Overall, our results provide further empirical support for the argument that assessing potential side effects of opaque algorithms is difficult (Busuioc, 2021; Gal et al., 2022). Our findings also support the view that various types of algorithmic transparency can reduce the likelihood of making discriminatory decisions, thereby enhancing the responsible use of these systems. This is in line with previous research that suggests a positive relationship between algorithmic transparency and reduction in algorithm-induced harms (e.g., Kordzadeh & Ghasemaghaei, 2022; M. K. Lee et al., 2019).

In addition, we illustrate that not all transparency mechanisms are effective in all situations; and demonstrate that handling different types of potentially discriminatory algorithmic recommendations requires different types of transparency. Indeed, transparency solutions that partially reveal the reasons for an algorithmic outcome, might mislead users to believing that the outcome is based on fair processes (Langer & König, 2023; Sokol & Flach, 2020a). Specifically, this study provides empirical evidence that when algorithmic recommendations include direct discrimination against a group (e.g., women), shedding light on the nature and contribution of each important variable used in producing recommendations (i.e., FI), is effective in increasing users' moral awareness and consequently reducing their tendency to readily approve potentially discriminatory recommendations. However, when the bias in recommendations is in

Table 5. The mediation role of users' perceived fairness of algorithmic recommendations that include indirect discrimination.

	Coeff.	SE	LLCI	ULCI
TG3 vs. TG5* → Perceived fairness	-.51	.21	-.93	-.08
Perceived fairness → Moral awareness	-.67	.09	-.85	-.50
TG3 vs. TG5* → Perceived fairness → Moral awareness	.34	.16	.04	.67

*TG was coded as a categorical variable where TG3 = 0 and TG5 = 1.

the form of indirect discrimination, providing FI was found to have no significant impact on users' moral awareness.

In fact, in this study, we hypothesised (H2a) that providing FI for recommendations biased by indirect discrimination can exacerbate the issue and reduce users' moral awareness. However, we did not find empirical support for this hypothesis. We believe that the reason for this could be that users tend to have low moral awareness (i.e., low levels of suspicion of the ethicality of system-generated recommendations) when they work with automated decision aids in general (Ebrahimi & Hassanein, 2019). Therefore, in the absence of compelling information/evidence that points them towards the possibility of biases in these systems' outputs, users are likely to consider algorithmic systems as generally unbiased (M. K. Lee, 2018). Similarly, the results of our post hoc analysis depict that receiving FI does not positively or negatively affect users' intention to approve indirectly discriminatory recommendations.

As such, we suggested, and our results confirmed that accompanying FI with ADI can alleviate the issue of users' adoption of recommendations contaminated by indirect discrimination. ADI, an approach to increasing algorithmic transparency suggested in this study, provides information about the distribution of each variable for different demographic groups. Therefore, it can elucidate whether the inclusion of one or more of these variables in a model has an adverse effect on a specific demographic group (e.g., women). Our results reveal that providing FI along with ADI in the context of indirectly discriminatory recommendations increases users' moral awareness and reduces the likelihood of their approval of the recommendations. These findings corroborate those of previous studies, which have also suggested that demographic explanations of system outputs can impact users' perceptions and cognitions (e.g., Ochmann et al., 2024; Shulner-Tal et al., 2022).

In addition, given the association between the transparency of algorithmic decision aids and users' trust in them (Grimmelikhuijsen, 2023) as well as the importance of trust in the success of these systems, in this study, we examined how providing various types of transparency impacts users' trust in algorithmic systems. Our results confirm that when users, by utilising transparency information, find out that an algorithmic system has employed a sensitive variable in its analysis, their trust in the system decreases. These findings complement those from previous studies, which indicate that when users have high perceptions of an algorithm's transparency and fairness, they begin to trust it (Shin, 2020). Specifically, our results indicate that when the transparency of an algorithm shows traces of direct discrimination in its recommendations, trust in the system deteriorates. As shown in

our post hoc analyses, this impact is in fact mediated by users' fairness perceptions, meaning that algorithmic transparency, which shows traces of direct discrimination in systems' outcomes, negatively influences users' fairness perceptions, which in turn undermines their trust in the system.

However, and contrary to our hypothesis H2b, when indirect discrimination was present in the experimental algorithm's outputs and, therefore, FI demonstrated a set of neutral variables, participants' trust in the system did not improve upon receiving FI. We believe that the reason for this could be that in our study, users had access to the entire data set of all employees and all features used in the model; therefore, providing them with such FI did not significantly impact their perception of the transparency of the system. Similarly, and in contrast with our H3b, when the set of variables included in FI is neutral, accompanying it with ADI that indicates potential bias in the analysis did not reduce users' trust. We suggest that the reason for this observation could be that in such circumstances, users do not blame the system for the bias and attribute the fault to other sources such as data sets. Indeed, theories of blame judgments suggest that although responsibility and blame are pertinent to both intentional and unintentional violations, they are stronger for intentional violations (Malle, 2021). Similarly, in the context of algorithmic systems, in a series of experiments, Sullivan and Fosso Wamba (2022) found that users attribute higher blame judgments towards these systems when a violation is perceived to be intentional compared to when it is perceived to be unintentional.

Finally, in line with Rest's theory (1986) our results illustrate that users' moral awareness precedes their adoption of discriminatory recommendations of algorithms. By enabling users to recognise that such recommendations can affect the welfare of data subjects and conflict with ethical standards, appropriate algorithm transparency triggers users' moral awareness (Butterfield et al., 2000), which in turn reduces their adoption of biased recommendations. The negative impact of moral awareness on users' engagement in ethical behaviour (or refraining from committing unethical behaviour) has been indeed repeatedly observed in the business ethics literature (e.g., Gok et al., 2017; Roy et al., 2023). Similarly, our results confirm the observation in the extant body of literature regarding the positive impact of users' trust in an algorithmic system on their adoption of the system's recommendations (Kim et al., 2021).

6.2. Implications for research

Algorithmic transparency has been widely seen as one of the predominant best practices in combating ethical issues in algorithmic decision making. Indeed, the link

between algorithmic transparency and perceptions of fairness of algorithmic-based decisions seems straightforward. However, humans are subject to bounded rationality and have their own cognitive biases which we contend may make the relationship between transparency and fairness of algorithmic decision making more complex. Therefore, this study investigated how different levels of feature transparency may have differing impacts on users' perceptions and, ultimately, their behavioural intention to approve discriminatory algorithmic recommendations depending on the underlying type of discrimination. By doing so, this study presents several research implications.

First, we contribute to research with an improved understanding of the influences of algorithmic transparency, which reveals biases hidden in algorithms' outputs. Despite the importance of the subject of the responsible use of algorithmic systems, empirical studies on this subject are scarce (Mikalef et al., 2022). Most extant studies on algorithmic bias are either technical or conceptual (Kordzadeh & Ghasemaghahi, 2022) without empirically studying how users, who are the ultimate decision-makers, in most cases, can be aided in better examining recommendations generated by algorithmic systems. Indeed, after a comprehensive review of the algorithmic bias literature, Kordzadeh and Ghasemaghahi (2022) call for empirical studies to examine the relationship between algorithmic transparency, users' cognitions, and their behavioural responses towards algorithmic results. Responding to their call, we empirically study the impact of increasing the transparency of algorithmic systems via the inclusion of FI explanations in two different contexts of biased system recommendations: direct and indirect discrimination. In the former context, our results not only confirm findings of previous research (e.g., Nyarko et al., 2021; Woodruff et al., 2018) regarding the negative influence of increasing FI transparency on users' attitudes (i.e., moral awareness), but also provide empirical evidence of similar influences in terms of users' intention to adopt algorithmic recommendations contaminated by direct discrimination. Specifically, while previous research has indicated that transparency that reveals algorithmic bias negatively impacts users' perceived fairness of algorithms and their recommendations (M. K. Lee et al., 2019; R. Wang et al., 2020), we provide empirical support that these perceptions of unfairness then cause a decline in users' adoption of biased recommendations, thereby bringing about a lower likelihood of biased algorithm-based decisions. As such, while prior research has provided empirical evidence for the positive impact of FI transparency on a variety of outcomes—such as improving users' understanding of algorithms (Štrumbelj & Kononenko, 2014), predicting how an algorithm would behave on unseen instances (Ribeiro et al., 2018), and improving users

over- or under-adhering to algorithmic recommendations (Balakrishnan et al., 2022), the findings of this study extend our understanding of the circumstances in which FI can decrease the incidence of discriminatory decisions.

Second, this study proposes and empirically tests the effectiveness of ADI, a new type of transparency, which when accompanied with FI helps users identify indirect discrimination when using algorithmic decision aids. Previous research suggests that technical transparency should be accompanied by transparency about the organisational, and decision-making contexts to better empower users when making complex decisions (Ehsan et al., 2021). By illustrating whether and how a feature(s) in a data set is (are) distorted in favour/against a certain demographic group, ADI enables users to judge whether the feature(s) is (are) inappropriate for the decision context, thereby causing a decline in the likelihood of adopting recommendations with indirect discrimination. Therefore, this study extends the extant body of literature on algorithmic decision-making by proposing a new type of transparency that reveals ethical issues in data sets used to train algorithms and/or generate recommendations, a concern expressed in the extant literature (Narayanan et al., 2024; Shah et al., 2023). In addition, by drawing upon Rest's theory of ethical decision making (Rest, 1986), we elucidate how providing transparency appropriate to the context and type of bias at hand can improve users' ethical decision-making, an approach categorised as enhancing users' actions towards an ethical decision by Poszler and Lange (2024). Indeed, the majority of the extant literature on ethical concerns of algorithmic decision-making (e.g., bias and accountability) assumes these issues as ethically salient and does not engage in a structured interrogation of the theoretical principles that guide such determination (Gal et al., 2022). The present study, therefore, takes a step towards addressing this issue by employing Rest's theory, and showing how the right type of transparency can influence users' fairness perceptions, enhance their awareness of ethical ramifications of the decision at hand, and consequently sway their decisions in the right direction ethically.

Third, our results concur with and extend previous research in the organisational science literature (e.g., Cropanzano et al., 2001; Folger & Konovsky, 1989; Sweeney & McFarlin, 1993), which suggests that procedural (un)fairness affects employees' attitudes towards events, authorities, and future behaviours (e.g., organisational commitment). By empirically demonstrating that users' observation of lack of fairness in the recommendations of an algorithm impacts their attitudes towards the decision at hand (i.e., moral awareness) as well as their behavioural intentions, our findings corroborate those of the organisational science literature (e.g., Konow, 2009; Searle et al., 2011; Taneja et al., 2023), in the context of users' observation of algorithms

engaging in procedurally unfair behaviour. In addition, our results extend previous findings that focused on human-human and employee-organisation relationships to human-intelligent decision aids relationships. Specifically, we empirically show that observations of procedural injustice on the part of algorithmic systems as responsible “powerful aids to decision-making” (Davenport et al., 2010, p. 17) have consequences in terms of users’ attitudes as well as their acceptance of algorithmic advice.

Fourth, our results explain the effects of identifying anomalies in the data analysis of algorithmic systems, which can induce direct and indirect discriminatory outcomes (Romei & Ruggieri, 2014) on users’ trust in the system. While the extant IS literature has mostly found a positive association between increased transparency of systems and users’ trust (e.g., Cramer et al., 2008; Sinha & Swearingen, 2001; W. Wang & Benbasat, 2007), the literature is less clear when it comes to empirically examining the relationship between the transparency of algorithmic systems and users’ trust (Mikalef & Gupta, 2021). Specifically, while some studies have suggested positive (Bedué & Fritzsche, 2022; Bitzer et al., 2021) and negative (Lehmann et al., 2020) relationships between these variables, some suggested that transparency can act as a double-edged sword in terms of its impact on users’ trust in an algorithmic system (Yu & Li, 2022). While previous research has provided different explanations for the apparent contradiction such as algorithmic complexity (Lehmann et al., 2020) and providing too much detailed information (Kizilcec, 2016), our findings confirm that finding an anomaly in the variables employed by an algorithm (i.e., direct discrimination) can hamper users’ trust in the system. However, producing a recommendation with indirect discrimination that is based on an apparently neutral set of variables does not have a similar impact on users’ trust. This study, therefore, contributes to the literature on trust in algorithmic systems and provides a partial answer to Mikalef and Gupta’s (2021, p. 260) call for studies that examine “the impact of transparency on use and trust formation in AI systems”. In addition, juxtaposing our findings with the ones provided by Newman et al. (2020), who suggest that increasing the transparency of decision processes improve users’ trust in a human decision maker but not in an algorithmic system, reveals that while users hold algorithmic decision aids accountable for the justice of their recommendations, the standards used to judge human decision makers and algorithms are not identical.

6.3. Implications for practice

Our study informs public and private organisations, which are increasingly employing algorithmic

decision aids in their decision-making processes, about the vulnerability of their employees to accepting potential biases embedded in the recommendations of these decision aids. Transparency, both in organisational practices and use of technology in decision making, is important and necessary to keep organisations accountable and help prevent problematic outcomes. Being able to detect and correct undesirable algorithmic behaviour, such as discriminatory recommendations or decisions, is critical to reduce reputational, commercial, and legal risks. While algorithmic transparency plays a central role to this detection, its effective application remains complex and situation specific. Organisations must be cautioned that the adage of “one size fits all” does not apply to transparency in algorithmic decision making (Sokol & Flach, 2020b). For instance, as our findings indicate, while feature importance is an intuitively simple transparency-increasing mechanism, it is not a reliable tool to detect algorithmic bias when it is in the form of indirect discrimination. FI alone is incapable of portraying a complete picture of sociotechnical algorithmic fairness issues, which stem from inequalities in societies and prior prejudicial decisions in companies and industries (Narayanan et al., 2024; Stapleton et al., 2022). Thus, identifying the presence and source of bias in algorithmic recommendations requires more careful examinations of the data and processes involved in generating algorithmic output.

Designers of algorithmic decision aids need to strike a delicate and difficult balance between simplicity and complexity of algorithmic transparency approaches. Too much information or complexity in transparency may create cognitive burden and result in an erosion of trust and fairness perceptions (Bitzer et al., 2023). Designers should utilize intuitively simple approaches to convey algorithmic transparency so that they are easily understood and appropriately considered by humans with varying skill sets and expertise under varying conditions of use and discriminatory sources. To overcome the shortcomings of the FI transparency method while maintaining an intuitively simple and accessible approach, we suggest algorithmic decision aid developers to augment FI with demographic-based measures to empower users in their endeavours to detect and mitigate algorithmic bias. Indeed, by empirically clarifying the role of transparency in triggering users’ moral awareness and eventually in their overriding of biased algorithmic recommendations, this study informs practitioners who strive for reducing instances of biased algorithmic decisions in their companies about the important role of more detailed transparency measures. Although some might argue that transparency can hurt companies due to the possibility of some users gaming the system (Wang et al., 2023), our results confirm that transparency

which enables users to examine the fairness of algorithmic outputs, improves users' compliance with ethical standards. This is especially important for more complex tasks because although the assistance of algorithms can improve procedural fairness in such tasks (Nagtegaal, 2021), highly complex tasks can overwhelm users and therefore cause a higher rate of users' approval of biased algorithmic recommendations (Ebrahimi & Matt, 2023). Therefore, our findings indicate that providing more fine-grained demographic transparency measures that enable users to probe algorithmic outputs in relation to unfairness towards (a) demographic group(s) is an asset for improving accountable use of algorithmic systems.

To this end, in this study, we proposed and empirically validated ADI, a means for more responsible use of algorithms, which reduces the likelihood of discriminatory algorithmic decision-making without compromising individuals' privacy. Indeed, ADI does not reveal sensitive features about individual data subjects (e.g., sex, race) and only presents this information in an aggregated form. Additionally, given that the suggested approach reveals variable(s) in data sets that can engender indirectly discriminatory outputs, it empowers practitioners by providing a clearer picture of decisions and, thus, in taking steps to tackle the issue. Examples for such steps include but are not limited to removing the problematic variable(s) from data sets (Robert et al., 2020) and examining the approach adopted to measure the variable(s) (Suresh & Guttag, 2021).

We found that users' trust in an algorithmic decision aid decreases when they find out that the system has used a sensitive variable such as data subjects' gender in generating a recommendation. Given the recent emphasis on improving practitioners' trust in and adoption of algorithmic systems, it is extremely important for developers of these systems to ensure that technical precautions are taken such that sensitive variables are prohibited from being employed in all data analysis. However, we did not find evidence of a similar impact when users found traces of indirect discrimination in algorithmic recommendations caused by biased data sets. These results, along with previous research that has found positive effects of increasing the transparency of complex systems such as algorithmic decision aids, bolster the importance of providing both common transparency mechanisms as well as demographic-related transparency mechanisms as proposed in this study. As such, this study benefits the developers of algorithms who not only seek to gain a competitive advantage by making their tools practical and more attractive to organisational users, but also might be liable for algorithm-based discriminatory decisions made by the users of their systems.

6.4. Limitations and future research

Despite the important contributions to research and practice outlined above, this study has several limitations that should be acknowledged. First, the experiment's participants were all middle managers employed in the US or Canada. Given the possible impact of culture on users' interactions with IT (Leidner & Kayworth, 2006) and more importantly, the impact of culture and varying regulations on individuals' understanding of discrimination and its unethicity as well as its consequences, our results must be extrapolated with caution to other countries and cultures. Additional studies conducted in different countries will help to increase the generalisability of the results of this study. In addition, given the importance of individuals' beliefs in dominant stereotypes in societies (e.g., prejudice against female employees), future research is required to explore how varying levels of such beliefs can impact the findings of this study.

The second limitation is related to the nature and characteristics of the decision participants were asked to make (i.e., selecting employees to be awarded the opportunity of attending a leadership training programme). Future research is warranted to conduct similar research in the context of different decisions to assess the validity of our results in different decisional contexts as previous research has shown that individuals' perceptions of the fairness of algorithms are impacted by the context such as promotion decisions being perceived as fairer than layoff decisions (Newman et al., 2020). Furthermore, the recommendations generated by the experimental tool in this study included discrimination against women. We suggest that future research to replicate and extend the findings of this study to contexts in which other demographic characteristics-such as race, age, or religion-are the basis for discriminatory recommendations. In addition, in this study the contribution of the sensitive variable (i.e., sex) was set to 20%. Future research is required to study whether different contribution levels of sensitive variables (e.g., 10% vs. 20% vs. 40%) would have a different impact on users' perceptions and behaviours. Moreover, since recent research has acknowledged the importance of task complexity on users' trust in algorithmic decision aids (Fan et al., 2020) as well as their cognitions and behaviours in the context of biased algorithmic recommendations (Ebrahimi & Matt, 2023), another fruitful path for future research can be examining the influences and efficacy of various transparency approaches in the context of direct and indirect discriminatory recommendations subject to various levels of task complexity.

Finally, this study was conducted in a context in which participants used the experimental tool (i.e., Smart HR-Tech) for the first time. Previous research

has shown that familiarity with a decision aid can impact the extent and outcome of users' interaction with the system and acceptance of its recommendations (Komiak & Benbasat, 2006; Sanders & Courtney, 1985). Familiarity can increase or decrease trust based on whether the trustor's interactions with the trustee are positive or negative (Luhmann, 2018). Indeed, Individuals' trust in technology often decreases over time as a result of facing various types of issues and error (De Visser et al., 2016; Madhavan & Wiegmann, 2007). However, this could be the opposite for novel technologies such as artificial intelligence where users might be sceptical of the maturity of such technologies (Glikson & Woolley, 2020). As such, future research is required to study the impact of transparency-increasing mechanisms proposed in this study on users' cognitions and behaviours in contexts where users are familiar with the algorithmic tool in question.

7. Conclusion

Building upon the literature on algorithmic bias, algorithmic transparency, and individuals' ethical behaviour, we theoretically and empirically studied users' cognitions and behavioural intentions when they receive transparency-increasing information that enables them to identify traces of potential direct and indirect discrimination in the recommendations of algorithmic systems. By doing so, we addressed an important gap in the extant literature in terms of conducting empirical studies related to users' cognitions and behaviours in the face of algorithmic bias. By studying the impact of FI—a common approach for increasing the transparency of algorithms—on reducing instances of users' approval of recommendations with direct and indirect discrimination, this study shed light on the circumstances in which such transparency-increasing approaches are helpful in combatting discrimination. Additionally, we proposed a novel transparency approach (i.e., ADI) as a complement to FI, which can reduce the likelihood of users' approval of algorithmic outputs with indirect discrimination.

Notes

1. Consistent with much prior ethical decision-making research (e.g., Haines et al., 2008; T. M. Jones, 1991), we consider the words moral and ethical synonyms and use them interchangeably.
2. Throughout the rest of this paper and unless otherwise stated, the terms “discriminatory recommendations” or “biased recommendation” refer to “potentially discriminatory recommendations” or “potentially biased recommendations”, respectively.
3. We did not study a treatment group that receives FI + ADI in the context of direct discrimination because, given the nature of direct discrimination (i.e.,

discrimination being based on a protected attribute such as gender) and in line with our H1, the ethical issue with a directly discriminatory recommendation can be found using only FI.

4. H1a, H1b, H2a, H2b, H3a, and H3b were also tested using MANOVA and t-test analyses and the results led to the same conclusions as the regression analyses. Data about these analyses are available upon request.
5. These post hoc analyses were only conducted for the hypothesised relationships that are supported by data. Specifically, we conducted these tests for H1a, H1b, and H3a.

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Appendix A

Table A1. Measurement instruments.

Constructs	Items
<i>Moral awareness</i>	
Please answer the following questions about the decision you were asked to make (about what employees to send to the leadership training programme)	
AWR1	The decision involved an ethical problem
AWR2	I was faced with an ethical issue
AWR3	The decision posed ethical issues
AWR4	I had to consider ethical issues in the decision
<i>Trust in algorithmic decision aid</i>	
TRST1	I can trust Smart HR-Tech
TRST2	I trust the information presented by Smart HR-Tech
TRST3	I trust the analysis performed by Smart HR-Tech
<i>Intention to approve recommendations</i>	
Given Smart HR-Tech's recommendation of the selected employees to be sent to the leadership training programme:	
APP1	I intend to follow it
APP2	I predict I would act on it
APP3	I plan to follow it
<i>Impression management</i>	
IM1	I sometimes tell lies if I have to (R).
IM2	I never cover up my mistakes.
IM3	There have been occasions when I have taken advantage of someone (R).
IM4	I always obey laws, even if I'm unlikely to get caught.
IM5	I have done things that I don't tell other people about (R).
IM6	I have said something bad about a friend behind his/her back (R).
<i>Reflective moral attentiveness</i>	
RMA1	I regularly think about the ethical implications of my decisions.
RMA2	I think about the morality of my actions almost every day.
RMA3	I often find myself pondering about ethical issues.
RMA4	I often reflect on the moral aspects of my decisions.
RMA5	I like to think about ethics.
<i>Perceived fairness of recommendations</i>	
FAIR1	The recommendation provided by Smart HR-Tech were fair.
FAIR2	The recommendations about employees to be sent to the training programme appeared to be fair.
<i>Manipulation Check (FI transparency)</i>	
FIT1	It was easy for me to understand the contribution of each variable in generating the recommendation provided by Smart HR-Tech.
FIT2	The degree of importance of the variables used in assessing candidates was clear to me.
<i>Manipulation check (ADI transparency)</i>	
ADIT1	I could easily see if the averages of the variables Smart HR-Tech used were different between men and women
ADT2	I could easily see if the averages of variables important in generating the recommendations varied between men and women.

Appendix B

Preliminary Data Analyses

Table B1. Construct reliability and discriminant validity.

	Mean	CA	AWR	TRST	APP
Moral Awareness (AWR)	3.75	0.91	0.85		
Trust in algorithmic decision aid (TRST)	5.36	0.91	−0.41	0.88	
Intention to Approve recommendations (APP)	5.08	0.94	−0.53	0.71	0.91

Diagonal elements are the square root of AVE; CA: Cronbach's alpha.

Table B2. Factor loadings and t-statistics (convergent validity).

Indicator	Loading	p-value	Indicator	Loading	p-value
<i>Moral awareness</i>			<i>Trust in algorithmic decision aid</i>		
AWR1	0.86	—	TRST1	0.89	—
AWR2	0.91	$p < 0.001$	TRST2	0.84	$p < 0.001$
AWR3	0.87	$p < 0.001$	TRST3	0.89	$p < 0.001$
AWR4	0.74	$p < 0.001$	<i>Intention to approve recommendations</i>		
			APP1	0.92	—
			APP2	0.88	$p < 0.001$
			APP3	0.94	$p < 0.001$